

Akshat S. Chaturvedi

Curriculum Vitae

Georgia State University Department of Physics and Astronomy

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EDUCATION

Georgia State University

PhD Astronomy (expected)

Atlanta, GA

2024 – Present

Georgia State University

MS Physics w/ Astronomy Concentration

Atlanta, GA

2024 – 2026

Pennsylvania State University

BS Astronomy & Astrophysics; BS Physics; Minor in Mathematics

University Park, PA

2020 – 2024

RESEARCH EXPERIENCE

Georgia State University

Graduate Research Assistant

Advisor: Dr. Douglas R. Gies

Atlanta, GA

Fall 2024 – Present

Project: *Investigating the fundamental properties of Be+sdOB binaries*

Research Expertise: Be stars, UV and optical spectroscopy, radial velocity analysis, cross correlation function analysis, Doppler tomography, long-baseline optical/near-IR interferometry

Pennsylvania State University

Undergraduate Research Assistant

Advisors: Dr. Robin Ciardullo and Dr. Howard E. Bond

University Park, PA

Fall 2023 – Summer 2024

Project: *A spectroscopic study of the peculiar planetary nebula Abell 57*

Research Expertise: Planetary nebulae, optical spectroscopy, photoionization modeling, UV photometry

Pennsylvania State University

Undergraduate Research Assistant

Advisors: Dr. Howard E. Bond, Dr. Robin Ciardullo, and Dr. Gautam Nagaraj

University Park, PA

Fall 2022 – Present

Project: *A census of blue post-AGB stars in Galactic globular clusters*

Research Expertise: Post-AGB stars, globular clusters, optical photometry, clustering algorithms, stellar populations, astrometric analysis

Pennsylvania State University

Undergraduate Research Assistant

Advisors: Dr. Kevin Luhman

University Park, PA

Summer 2022

Project: *A chrono-kinematic analysis of nearby star associations*

Research Expertise: Clustering algorithms, stellar populations, kinematic analysis, traceback age analysis

PUBLICATIONS

Detection and Astrometry of the Ba–Bb Subsystem in α Piscium: First Dual-field Interferometry at the CHARA Array

N. Anugu, R. Klement, [...] **A. S. Chaturvedi**, et al. 2026

The Astronomical Journal, Volume 171, Issue 4, id. 253

Broad-band Monitoring of 797 Montana

M. C. Bentz, R. Alakhone, L. Azoulay, E. Burns-Kaurin, **A. S. Chaturvedi**, et al., 2025

The Minor Planet Bulletin Vol. 52, No. 3, pp. 207-208

R-Band Monitoring of Patroclus and Menoetius Mutual Events

M. C. Bentz, R. Brown, S. Carrasco-Gaxiola, **A. S. Chaturvedi**, et al., 2025

The Minor Planet Bulletin Vol. 52, No. 1, pp. 79-81

Spectroscopic Survey of Faint Planetary-nebula Nuclei. V. The EGB 6-type Central Star of Abell 57

H. E. Bond, **A. S. Chaturvedi**, et al., 2024

The Astrophysical Journal, Volume 970, Issue 2, id.164

CONFERENCE TALKS AND POSTERS

42nd Liège Astrophysical Colloquium

Talk: Masses and Properties of Be+sdO/B Binaries From 3D Orbits

Liège, Belgium

July 2026

2026 CHARA/MROI Science Meeting

Talk: Exploring the Binarity of Be Stars

Socorro, NM

March 2026

Georgia Regional Astronomy Meeting

Talk: Be Stars and Beyond: A Legacy of Binary Mass Transfer

Atlanta, GA

November 2025

Staraganza III

Talk: Be Stars and Beyond: A Legacy of Binary Mass Transfer

Atlanta, GA

September 2025

Galaxies to Gluons Summer Lunch Talk

Talk: Be Stars and Beyond: A Legacy of Binary Mass Transfer

Atlanta, GA

July 2025

Penn State Undergraduate Research Exhibition

Poster: A Spectroscopic Study of the Peculiar Planetary Nebula Abell 57

University Park, PA

April 2024

American Physical Society Mid-Atlantic Section Meeting

Poster: A Census of Blue Post-Horizontal-Branch Stars in Galactic Globular Clusters

University Park, PA

Nov 2025

AWARDS, FUNDING, AND SCHOLARSHIPS

Georgia State University Visiting Observers Grant

\$1400 to support travel to the 42nd Liège Astrophysical Colloquium

Georgia Space Grant Consortium

April 2026

Travel Support Award <i>\$850 to support travel and lodging</i> 2026 CHARA/MROI Science Meeting and New Visions in Optical Interferometry Workshop	March 2026
Second Century Initiative (2CI) Fellow Georgia State University	Fall 2024 – Present
Faculty Undergraduate Research Support Award <i>\$1000 to support publication costs</i> Pennsylvania State University	Spring 2024
Induction into Sigma Pi Sigma <i>National physics and astronomy honors society</i> Pennsylvania State University	Spring 2024
Induction into Phi Beta Kappa <i>National liberal arts and sciences honors society</i> Pennsylvania State University	Spring 2023
Eberly College of Science Dean's List Pennsylvania State University	2021, 2022, Fall 2023, and Spring 2024
George E., Jr. and Elizabeth S. Sperling Tutorial Endowment Scholarship <i>\$1000 awarded to learning assistants who excel in tutorial ability</i> Pennsylvania State University	Spring 2022

OUTREACH AND LEADERSHIP

President Physics Graduate Student Association, GSU Organization for advocating for physics and astronomy graduate students at GSU	Atlanta, GA Fall 2026 – Spring 2027
Georgia State University Undergraduate Research Conference — Judge Graduate student judge for physics and astronomy posters	Atlanta, GA April 2026
Astronomy on Tap Atlanta — Speaker Talk: Final Destination: The Epic Ends of Stellar Lives Viewed by Next Generation Telescopes	Atlanta, GA August 2025
AsTropaLooza Citizen Science Workshop – Volunteer Astronomy of Atlanta K–12 STEM outreach event	Atlanta, GA June 2025
Department of Physics & Astronomy Observing Nights – Volunteer Observing nights for astronomy students	Atlanta, GA Spring 2025 – Present
Hard Labor Creek Observatory Public Open Nights – Volunteer Public open house at Georgia State University's Hard Labor Creek Observatory	Rutledge, GA Fall 2024 – Present
Vice President Society of Physics Students Pennsylvania State University	University Park, PA Fall 2023 – Spring 2024
Society of Physics Students CV Building Workshop – Lead Pennsylvania State University	University Park, PA Fall 2023

OBSERVING EXPERIENCE

Astrophysical Research Consortium 3.5m Telescope

Apache Point Observatory
Sunspot, NM

— *Obtained high-resolution echelle spectra of high- and intermediate-mass stars using the ARCES instrument*

December 2025 – Present

Observing Time: 52 hours

Center for High Angular Resolution Astronomy (CHARA) Array

Mount Wilson Observatory
Mount Wilson, CA

— *Obtained long-baseline near-infrared interferometric observations of Be+sdOB stars*

July 2025 – Present

Observing Time: 32 hours

Miller 24" Telescope

Hard Labor Creek Observatory
Rutledge, GA

— *Obtained photometric lightcurves of the asteroids 617 Patroclus, Menoetius, 797 Montana, and 4107 Rufino*

Fall 2024 – Spring 2025

Observing Time: 50 hours

Department of Astronomy & Astrophysics 24" Telescope

Pennsylvania State University
University Park, PA

— *Obtained photometric observations and images of a variety of astronomical objects, including planetary nebulae, globular clusters, and planets*

Fall 2022

Observing Time: 20 hours

TEACHING EXPERIENCE

Astronomy 1020K | Stellar and Galactic Astronomy (x9)

Georgia State University
Graduate Teaching Assistant

Fall 2024 – Present

Atlanta, GA

Astronomy 1010K | Astronomy of the Solar System (x1)

Georgia State University
Graduate Teaching Assistant

Spring 2025 – Present

Atlanta, GA

Astronomy & Astrophysics 292 | Astronomy of the Distant Universe

Pennsylvania State University
Undergraduate Learning Assistant

Spring 2023

University Park, PA

Astronomy & Astrophysics 291 | Astronomical Methods and the Solar System

Pennsylvania State University
Undergraduate Learning Assistant

Fall 2022

University Park, PA

Physics 212M | General Physics: Electricity and Magnetism

Pennsylvania State University
Undergraduate Learning Assistant

Spring 2022

University Park, PA

Physics 212M | General Physics: Mechanics

Pennsylvania State University
Undergraduate Learning Assistant

Fall 2021

University Park, PA

PROFESSIONAL MEMBERSHIP

PLATO Multiple Star Working Group

Complementary science working group for ESA's PLATO mission

2025–Present

American Astronomical Society

American society of professional astronomers

2024–Present

SOFTWARE

CATCH (CHARA Array's Thrifty Calibrator Hunter)

A tool to find and verify calibrator stars for interferometry targets at the CHARA Array

TECHNICAL SKILLS & LANGUAGES SPOKEN

Programming: Python, ADQL, R, Julia, Rust

Software: TOPCAT, CLOUDY, AstroImageJ, SAO DS9, IRAF

Languages: English (native), Hindi (native), Marathi (native), French (A2)